Deep Learning for Natural Language Processing

Today's goal • 3/3 items

Module 1
Feedforward Neural Nets and Recurrent Neural Networks

Module 2
Sequence to Sequence Models, Attention, Transformers

Recurrent Sequence-to-Sequence Models and the First Attention Mechanism

- Sequence-to-sequence Tasks in NLP
  Video • 5 min
- Early Recurrent Sequence-to-Sequence Models
  Video • 6 min
- Alignment in Machine Translation
  Video • 4 min
- Sequence-to-Sequence Models with Attention
  Video • 5 min
- Sequence-to-Sequence Models and Attention
  Practice Assignment • Grade: 100%

Transformer Models

- The Transformer Model
  Video • 7 min
- Mathematics of Multi-Head Attention
  Reading • 40 min
- Applications of the Transformer Model
  Video • 6 min
- Variants of the Attention Mechanism
  Video • 6 min
- Transformer Models
  Practice Assignment • Grade: 100%

Tips and Tricks for Training Neural Networks

- Regularization
  Video • 8 min
- Optimizers
  Video • 4 min
- Efficient Model Training
  Video • 5 min
- Tips and Tricks for Training Neural Networks
  Practice Assignment • Grade: 100%

Review and Practice

- Sequence-to-sequence Models, Attention, and Transformers
  Graded Assignment • Grade: 100%
- **Transformers for Sequence Classification**
  Programming Assignment • Grade: 100%

Module 3



Programming assignment

Transformers for Sequence Classification

Launch lab

What to expect

Due Nov 13, 11:59 PM IST

Unlimited attempts

You passed!
To pass you need at least 80/100 points.

100/100

Instructions My submissions Discussions

In this assignment, you will build and train a neural network for text classification using PyTorch. You will start by preprocessing text data, constructing a vocabulary, and encoding sentences into numerical representations. Using this processed data, you will implement key components of a Transformer-based model, including positional encoding, multi-head attention, and a feedforward network.

Through this assignment, you will:

- Develop a text preprocessing pipeline, including tokenization and vocabulary construction.
- Implement custom dataset classes to handle text data efficiently in PyTorch.
- Gain a deeper understanding of self-attention mechanisms and their role in modern NLP architectures.
- Train and evaluate a Transformer-like model for spam classification.
- Explore the impact of key hyperparameters on model performance.
- By the end, you will have hands-on experience with text-based deep learning, helping you understand how Transformers process and classify textual inputs.

Report an issue

Go to next item →